



SKIPS University – School of Computer Science

Integrated MSc – Information Technology; Batch 2024–29

Bachelor of Computer Applications (Hons.); Batch 2024–28

Semester – III

Practical Session Plan

Course Code : SOCS030303 / SOCS020303

Course Title : Database Management System

Credit : 4

Credit Hours : 75 hours (45 Hours Theory + 30 Hours Lab)

Instructor : Ms. Shruti Malde

E-mail ID : shruti.m@skipsuniversity.edu.in

Session No.	Practical																																
1	Installation of Oracle - Connecting Client to Oracle Server Post Installation Steps Basic Table Creation																																
	<table><tr><th>Table Name</th><th>Columns & Data Types</th></tr><tr><td rowspan="3">Department</td><td>Dept_ID – Number(5)</td></tr><tr><td>Dept_Name – Varchar2(50)</td></tr><tr><td>Location – Varchar2(30)</td></tr><tr><td rowspan="5">Employee</td><td>Emp_ID – Number(6)</td></tr><tr><td>Emp_Name – Varchar2(50)</td></tr><tr><td>Job_Title – Varchar2(30)</td></tr><tr><td>Salary – Number(8,2)</td></tr><tr><td>Dept_ID – Number(5)</td></tr><tr><td rowspan="5">Student</td><td>Student_ID – Number(6)</td></tr><tr><td>First_Name – Varchar2(30)</td></tr><tr><td>Last_Name – Varchar2(30)</td></tr><tr><td>DOB – Date</td></tr><tr><td>Gender – Char(1)</td></tr><tr><td rowspan="4">Course</td><td>Course_ID – Varchar2(10)</td></tr><tr><td>Course_Name – Varchar2(50)</td></tr><tr><td>Duration – Number(2)</td></tr><tr><td>Fees – Number(8,2)</td></tr><tr><td rowspan="5">Book</td><td>Book_ID – Number(5)</td></tr><tr><td>Title – Varchar2(100)</td></tr><tr><td>Author – Varchar2(50)</td></tr><tr><td>Price – Number(6,2)</td></tr><tr><td>Publication_Year – Number(4)</td></tr><tr><td rowspan="3">Library_Member</td><td>Member_ID – Number(6)</td></tr><tr><td>Member_Name – Varchar2(50)</td></tr><tr><td>Join_Date – Date</td></tr></table>	Table Name	Columns & Data Types	Department	Dept_ID – Number(5)	Dept_Name – Varchar2(50)	Location – Varchar2(30)	Employee	Emp_ID – Number(6)	Emp_Name – Varchar2(50)	Job_Title – Varchar2(30)	Salary – Number(8,2)	Dept_ID – Number(5)	Student	Student_ID – Number(6)	First_Name – Varchar2(30)	Last_Name – Varchar2(30)	DOB – Date	Gender – Char(1)	Course	Course_ID – Varchar2(10)	Course_Name – Varchar2(50)	Duration – Number(2)	Fees – Number(8,2)	Book	Book_ID – Number(5)	Title – Varchar2(100)	Author – Varchar2(50)	Price – Number(6,2)	Publication_Year – Number(4)	Library_Member	Member_ID – Number(6)	Member_Name – Varchar2(50)
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403	PL/SQL Programming	Steven Feuerstein	550.00	2019
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Course Table

Course_ID	Course_Name	Duration (Months)	Fees
C101	Database Systems	4	15000
C102	Data Structures	3	12000
C103	Web Development	6	20000

3

Implement the following SQL Queries based on the given above table.

1. Write a query to add a new column Email of type VARCHAR2(50) to the Student table.
2. Write a query to add a column Publisher (VARCHAR2(50)) to the Book table.
3. Write a query to add two new columns to the Course table:
4. Start_Date (DATE)
5. Instructor (VARCHAR2(40))
6. Write a query to drop the column Salary from the Employee table.
7. Write a query to drop the column Location from the Department table.

4

8. Write a query to modify the size of the Emp_Name column in the Employee table to 100 characters.
9. Write a query to change the data type of Fees in the Course table to NUMBER (10,2).
10. Write a query to rename the column Course_Name to Title in the Course table.
11. Write a query to rename the table Student to Students.
12. Write a query to add a new column Membership_Fee (NUMBER(6,2)) to the Library_Member table.

5

1. Create a table named Faculty with the following columns and apply constraints:
 - Faculty_ID (Number) – Primary Key
 - Faculty_Name (Varchar2(50)) – Not Null
 - Department (Varchar2(30))
 - Email (Varchar2(50)) – Should be unique
2. Create a table named Subject with:
 - Subject_Code (Varchar2(10)) – Primary Key
 - Subject_Name (Varchar2(50)) – Not Null
 - Credits (Number) – Should be between 1 and 6
 - Faculty_ID (Number) – Foreign Key referencing Faculty (Faculty_ID)
3. Create a table called Registration having:
 - Reg_ID (Number) – Primary Key
 - Student_ID (Number) – Not Null
 - Course_ID (Varchar2(10)) – Not Null
 - Registration_Date (Date)
 - Add a unique constraint on the combination of Student_ID and Course_ID
4. Create a table named Login with the following structure and constraints:
 - Username (Varchar2(20)) – Primary Key
 - Password (Varchar2(20)) – Not Null

	• Role (Varchar2(15)) – Must be either 'Admin', 'Student', or 'Faculty' • Status (Char(1)) – Default value should be 'A'																																
6.	5. Create a table Exam_Result with: • Result_ID (Number) – Primary Key • Student_ID (Number) – Foreign Key referencing Student(Student_ID) • Subject_Code (Varchar2(10)) – Foreign Key referencing Subject(Subject_Code) • Marks_Obtained (Number) – Should be between 0 and 100 • Grade (Char(1)) 6. Create a table named Vendor with the following columns and constraints: • Vendor_ID (Number) – Primary Key • Vendor_Name (Varchar2(50)) – Not Null • City (Varchar2(30)) • Contact_Number (Varchar2(15)) – Must be unique • Rating (Number) – Must be between 1 and 5 7. Create a table Product with these details: • Product_ID (Number) – Primary Key • Product_Name (Varchar2(50)) – Not Null • Price (Number) – Should be greater than 0 • Quantity_Available (Number) • Vendor_ID (Number) – Foreign Key referencing Vendor(Vendor_ID) 8. Create a table named Attendance with the following structure: • Attendance_ID (Number) – Primary Key • Student_ID (Number) – Foreign Key referencing Student(Student_ID) • Attendance_Date (Date) • Status (Char(1)) – Should be either 'P', 'A', or 'L' • Add a unique constraint on (Student_ID, Attendance_Date)																																
7.	Insert Records into the appropriate tables. Faculty Table<table><tr><th>Faculty_ID</th><th>Faculty_Name</th><th>Department</th><th>Email</th></tr><tr><td>101</td><td>Nidhi Patel</td><td>IT</td><td>nidhi.p@university.edu</td></tr><tr><td>102</td><td>Raj Mehta</td><td>Finance</td><td>raj.m@university.edu</td></tr><tr><td>103</td><td>Sneha Shah</td><td>Marketing</td><td>sneha.s@university.edu</td></tr></table> Subject Table<table><tr><th>Subject_Code</th><th>Subject_Name</th><th>Credits</th><th>Faculty_ID</th></tr><tr><td>DBMS101</td><td>Database Systems</td><td>4</td><td>101</td></tr><tr><td>MKT102</td><td>Marketing Basics</td><td>3</td><td>103</td></tr><tr><td>FIN103</td><td>Financial Accounting</td><td>5</td><td>102</td></tr></table>	Faculty_ID	Faculty_Name	Department	Email	101	Nidhi Patel	IT	nidhi.p@university.edu	102	Raj Mehta	Finance	raj.m@university.edu	103	Sneha Shah	Marketing	sneha.s@university.edu	Subject_Code	Subject_Name	Credits	Faculty_ID	DBMS101	Database Systems	4	101	MKT102	Marketing Basics	3	103	FIN103	Financial Accounting	5	102
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FIN103	Financial Accounting	5	102																														

Login Table

Username	Password	Role	Status
user_2001	pass123	Student	A
admin_user	admin@123	Admin	A
f_mehta	fac456	Faculty	A

- ✓ Role values allowed: 'Admin', 'Student', 'Faculty'
- ✓ Status default is 'A'

Registration Table

Reg_ID	Student_ID	Course_ID	Registration_Date
1	2001	C101	2024-06-01
2	2002	C102	2024-06-05
3	2001	C102	2024-06-08

Insert Records into the appropriate tables.

Exam_Result Table

Result_ID	Student_ID	Subject_Code	Marks_Obtained	Grade
1	2001	DBMS101	85	A
2	2002	FIN103	68	B
3	2003	MKT102	90	A

Vendor Table

Vendor_ID	Vendor_Name	City	Contact_Number	Rating
V001	TechBooks	Mumbai	9876543210	5
V002	EduSupply	Delhi	9123456789	4
V003	PrintMart	Pune	9988776655	3

- ✓ Contact_Number is unique
- ✓ Rating must be between 1 and 5

8.

Product Table

Product_ID	Product_Name	Price	Qty_Available	Vendor_ID
P101	Oracle Textbook	500	50	V001
P102	Ledger Book	300	80	V002
P103	Marketing Handbook	450	60	V003

- ✓ Price > 0
- ✓ Vendor_ID references Vendor

Attendance Table

Attendance_ID	Student_ID	Attendance_Date	Status
1	2001	2024-07-01	P
2	2002	2024-07-01	A
3	2003	2024-07-01	L

- ✓ Status: Only 'P', 'A', 'L'
- ✓ Unique constraint on (Student_ID, Attendance_Date)

9**Implements the following SQL queries based on given suitable table.****Faculty Table**

1. Display all records from the Faculty table.
2. Display the names and emails of all faculty members.
3. List all faculty members from the "IT" department.
4. Find the faculty member whose email is 'nidhi.p@university.edu'.

Subject Table

5. Show the subject name and credits for all subjects.
6. Display details of all subjects taught by faculty with Faculty_ID = 101.
7. List subjects where the credits are greater than 3.

Registration Table

8. Display all records from the Registration table.
9. Show all course registrations made by student with Student_ID = 2001.
10. Find all students who registered for course C102.

10**Login Table**

11. Display usernames and roles from the Login table.
12. List all users with role 'Student'.
13. Show all active users (Status = 'A').

Exam_Result Table

14. Show all results where marks obtained are greater than 80.
15. List student IDs and grades for subject DBMS101.
16. Find students who got grade 'A'.

Vendor Table

17. List all vendors located in 'Mumbai'.
18. Display vendor names with a rating of 4 or higher.
19. Show contact numbers of all vendors.

11	Product Table 20. Display all products with a price greater than 400. 21. List all products supplied by vendor V001. 22. Show product names and quantities where quantity is less than 70. Attendance Table 23. Display all attendance records for 2024-07-01. 24. Find all students who were marked 'P' (Present). 25. Show the attendance status of student 2003.
12	26. List all attendance records in alphabetical order of status. 27. List all products sorted by price in ascending order. 28. Display all products in descending order of quantity available. 29. List all products sorted first by Vendor_ID and then by Price in ascending order. 30. List products with price more than 400, sorted by Product_Name. 31. Display all products from vendor V002 sorted by quantity available in ascending order.
13	32. Display the absolute difference between 75 and Marks_Obtained for each student. 33. List students with their Marks_Obtained, and display the ceiling of Marks_Obtained divided by 7. 34. Display Subject_Code in uppercase and lowercase. 35. Show the first 3 characters of each Subject_Name from the Subject table. 36. Display student IDs and their Grades in lowercase. 37. Convert the string '2024-07-11' to a proper date format. 38. Show registration dates in DD-MM-YYYY format for each student.
14	39. Display the day of the week for each attendance record. 40. Find the total marks obtained by all students. 41. Calculate the average marks obtained in all subjects. 42. Count how many students appeared in the exam. 43. Count the number of students who scored more than 70 marks. 44. Find the total marks obtained by each student.
15	Implement the Following Queries Based on the Given Tables- Exam_Result Table Subject Table Registration Table 45. Display student marks along with subject name by joining Exam_Result and Subject tables. 46. Show all registration records with their subject names, even if the subject is missing. 47. List all students who scored more than 80 marks along with subject names. 48. Find all subjects along with the faculty ID and students who took them in the exam. 49. Display student ID and their marks, along with the average marks of all students. 50. Find students who scored above the average marks.